

Granularity, Not Bit-Width: Group-Wise INT4 Preserves Generation Coherence of Driving-VLA Planners Under Deployment-Domain Shift

Justin Williams, Kishor Datta Gupta, Roy George, Mrinmoy Sarkar
Clark Atlanta University
justinwilliamstech693@gmail.com

Abstract—We study weight-only INT4 quantization of the language/action head of OpenDriveVLA-0.5B in a closed-loop neural-rendering simulator (NeuroNCAP, 14 real-camera scenes). The headline: under the deployment-domain shift induced by photorealistic re-rendering, naive per-output-channel INT4 catastrophically loses generation coherence—the planner emits degenerate trajectory text on nearly every frame and effectively freezes ($\leq 15\%$ well-formed on 15/16 scenario instances, mean 3.7%)—whereas group-wise INT4 (group size 128, an extra ≈ 0.1 bits/weight) recovers most of FP16’s coherence (mean 73.5% vs 76.0%, $\geq 85\%$ of FP16 on 13/16). The same per-channel INT4 is lossless and well-formed in-distribution (open-loop, real images), so the failure is a quantization $\times$ domain-shift interaction governed by scale granularity, not by 4-bit precision per se. We report generation-coherence rate as the clean metric and discuss why downstream collision/NCAP counts are a confounded, scene-dependent proxy (on one scene every config including FP16 collides).

Index Terms—vision-language-action models, autonomous driving, post-training quantization, group-wise INT4, closed-loop evaluation, generation coherence, edge deployment, Jetson Orin

I. Introduction

Group-wise (block-wise) weight quantization with per-group scales is the de-facto standard for INT4 LLMs because a single scale shared across many weights is set by the group’s largest magnitude, so one outlier inflates the quantization step for every weight it shares it with. We ask whether this granularity matters for the closed-loop behavior of a quantized driving VLA under the distribution shift of a neural-rendering deployment surrogate, and find that it is decisive.

II. Setup

We quantize only the 168 Linear layers of the Qwen2.5-0.5B backbone (357.8M parameters); perception (UniAD track+map), projectors, embeddings, and head stay FP. We run closed-loop on 14 NeuroNCAP scenes (16 scenario instances): a NeuRAD renderer re-renders six cameras at the ego’s driven pose each step, a model node returns a trajectory, and a kinematic controller advances the vehicle. Ten random actor-jitter seeds are run paired across precisions (jitter seeded by run index; only the quantization

differs). Our primary metric is generation coherence: the fraction of frames on which the planner emits a well-formed (non-degenerate) trajectory, computed from the logged per-frame plans.

III. Results

a) The effect is domain-shift-induced, not 4-bit precision.: On real nuScenes images (open-loop) the same per-channel INT4 is lossless (0.33m L2, well-formed). It collapses only on rendered (out-of-distribution) inputs, and this replicates across all 16 scenario instances ($\leq 15\%$ well-formed on 15/16). Group-wise INT4 is robust ($\geq 85\%$ of FP16 on 13/16, partial on 0099/0101/0796). The fix—finer scale granularity—costs $16/128 = 0.125$ extra bits per weight.

b) Mechanism.: A shared INT4 scale is $s = \max |w|/7$. One large-magnitude weight inflates s for the whole group, coarsening the grid for the small weights that carry signal. Group-wise scales confine each outlier to its own block. Under distribution shift the planner’s hidden states move; per-channel’s coarsened grid then tips greedy decoding into malformed trajectory text, while group-wise’s finer grid preserves coherence.

IV. On the metric

We deliberately report generation coherence rather than collision rate. The downstream collision outcome is a confounded, scene-dependent proxy: on some scenes per-channel INT4 collides more than group-wise/FP16, but on several scenes every config—including FP16—collides 10/10 because the base policy cannot clear that scenario. A collapsed (frozen) planner only collides on geometries that strike a stationary ego, and the count depends on the degenerate fallback. Coherence isolates the quantization effect; collisions do not.

V. Limitations

Results are on 14 scenes (16 scenario instances)—the full released closed-loop benchmark, not all of nuScenes; the rendered inputs are out-of-distribution to real nuScenes

TABLE I

Closed-loop generation coherence (well-formed predicted-frame rate), real-camera NeuRAD rendering, 14 scenes / 16 scenario instances, 10 paired seeds each.

Config	mean coherence	per-scenario behaviour
FP16	76.0%	—
INT4 per-channel (naive)	3.7%	collapses ($\leq 15\%$) on 15/16
INT4 group-128	73.5%	recovers ($\geq 85\%$ of FP16) on 13/16

(on several scenes even FP16 fails the collision test, 10/10), so the claim is the relative one (group $\approx$ FP16 $\gg$ per-channel on coherence). Group-wise recovery is partial on 3 scenes (0099, 0101, 0796) —we do not claim exact equivalence. Lower-bit (INT3/INT2) and activation-aware (AWQ) variants were characterized only in an earlier single-scene setup and are not re-reported here pending a real-camera re-run.

VI. Conclusion

For quantized driving-VLA planners under deployment-domain shift, scale granularity, not bit-width, governs whether generation stays coherent. Group-wise INT4 (≈ 0.1 extra bits/weight) matches FP16 robustness where naive per-channel INT4 collapses—an actionable, nearly-free deployment recommendation that open-loop L2 cannot motivate.

References

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